

System Analysis and Design

Project Requirement Outline

Introduction: In this project, you will design an information system to address a real-world problem. You will move through the phases of the Software Development Life Cycle (SDLC), focusing specifically on identifying the problem, gathering data, and analyzing system needs. You will then create several key diagrams to represent the flow of information and project timelines. Your goal is to present your findings and system design through clear visual diagrams.

Phase 1: Identifying the Problem

What is required:

- **Problem Definition:**
 - Clearly define the problem the system will address. Consider a real-world issue that could benefit from an information system.
 - Describe how the problem affects the stakeholders (e.g., users, businesses) and provide context for the system's development.
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Phase 2: Determining Information Requirements

What is required:

- **Conduct Interviews:**
 - Interview **at least 3-5 stakeholders** (users, business owners, or other relevant individuals) to understand their needs and expectations.
 - **Prepare questions** focusing on system functionalities, current challenges, and desired features.
 - **Interview Insights:**
 - Document the key information gathered, including the pain points, goals, and expectations for the system.

Phase 3: Analyzing System Needs

What is required:

- **System Requirements Analysis:**

- Based on the data from interviews, identify the key system requirements that address the problem.
- List the system's core features and functionalities (e.g., user login, data processing, reporting).

- **Required Diagrams:**

- **Context Flow Diagram (DFD):**

- Create a **Context Diagram** that shows how the system interacts with external entities (e.g., users, third-party systems).
- Illustrate how information flows between the system and these external entities.

- **Use Case Diagram:**

- Design a **Use Case Diagram** to represent the system's primary actors (e.g., users, administrators) and their interactions with the system.
- Identify and describe the key use cases (e.g., placing an order, generating reports).

- **Gantt Chart:**

- Create a **Gantt Chart** to outline the project timeline.
- Include key milestones, tasks, and estimated durations for each phase of the project.

- **PERT Chart:**

- Develop a **PERT Chart** to visualize the project's critical path.
- Show task dependencies and estimated durations to ensure the project is completed on schedule.

- **DFD Diagram 0 and Child Diagram:**

- Design a Level 0 Data Flow Diagram (DFD) to depict the system's main processes, inputs, and outputs.

- Develop child diagrams for more detailed processes to ensure a comprehensive understanding of the system's workings.
- Class Diagram:
 - Develop a Class Diagram that outlines the system's main classes, their attributes, and relationships.
- Activity Flow Diagram:
 - Create an Activity Diagram to represent the workflow of critical processes within the system.

Instructions:

Groups of 2-3 students

1. **Choose a real-world problem** that can be solved with an information system. The problem should be relevant and address a gap or inefficiency that can benefit from automation or improved processes.
2. **Conduct interviews** with at least 3-5 stakeholders to gather requirements for the system. Document the responses and insights.
3. **Analyze the gathered data** and define the system requirements.
4. **Design the required diagrams** (Context Flow Diagram, Use Case Diagram, Gantt Chart, and PERT Chart) to represent the system, its requirements, and the project timeline.
5. **Submit your project** including:
 - A brief report that explains the problem, your interviews, and your findings.
 - The required diagrams with explanations on how they contribute to solving the problem.
 - The Gantt and PERT charts that outline the project's milestones and timelines